

Computational Graph (CG)

Operator

$+$, $*$, $/$, $-$ binary

$\sin$, $\cos$, $\exp$ unary operator



Operand

x, y, z independent

t_1, t_2, t_3 dependent



operator

Forward pass $\frac{\partial F}{\partial x} = 0 + (-\sin(xy))y$

Reverse $\frac{\partial F}{\partial x} = 1$

$f(x, y, z) = \sin(yz) + \cos(xy) + e^{x+y}$

MIPS

$t_1 = y * z$

$t_2 = x * y$

$t_3 = x + y$

$t_4 = \sin(t_1)$

$t_5 = \cos(t_2)$

$t_6 = \exp(t_3)$

$t_7 = t_5 + t_6$

$t_8 = t_7 + t_4$

$\frac{\partial t_8}{\partial x} = \frac{\partial t_7}{\partial x} + \frac{\partial t_4}{\partial x}$

$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial t_8} \frac{\partial t_8}{\partial x}$

$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial t_8} \left(\frac{\partial t_7}{\partial x} + \frac{\partial t_4}{\partial x} \right)$

$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial t_8} \left(\frac{\partial t_5}{\partial x} + \frac{\partial t_6}{\partial x} + \frac{\partial t_4}{\partial x} \right)$

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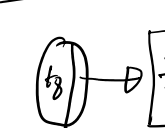
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Reverse



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$= -y \sin(xy) + e^{x+y}$

$f(x) = x^6$

$\frac{\partial F}{\partial x} = 6x^5$

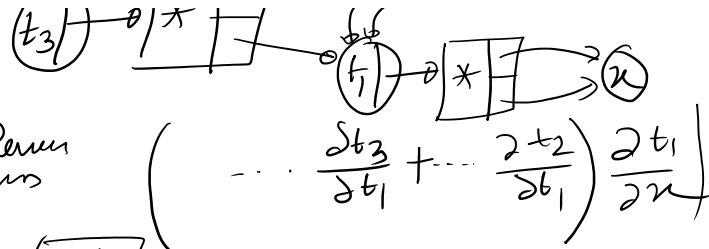
$t_1 = x * x$

$t_2 = x * t_1$

$t_3 = x * t_2$

$\frac{\partial L}{\partial u}$

$$\begin{aligned} t_1 &= x * x \quad // x^2 \\ t_2 &= t_1 * t_1 \quad // x^4 \\ t_3 &= t_1 * t_2 \quad // x^6 \end{aligned}$$



$$\begin{aligned} \frac{\partial t_3}{\partial u} &= \left(\frac{\partial t_3}{\partial t_3} \frac{\partial t_3}{\partial t_1} + \frac{\partial t_3}{\partial t_2} \frac{\partial t_3}{\partial t_2} \frac{\partial t_2}{\partial t_1} \right) \left(\frac{\partial t_1}{\partial u} \right) \\ &= (1 \cdot t_2 + 1 \cdot t_1 \cdot 2t_1) (2x) \\ &= (x^4 + 2x^4) 2x \\ &= 2x^5 + 4x^5 \\ &= 6x^5 \end{aligned}$$

$5x^4$

$$\begin{aligned} t_1 &= x * x \\ t_2 &= t_1 * t_1 \\ t_3 &= t_2 * x \end{aligned}$$

- Perceptron
- MLP
- CG
- Process Training
 - Optimization
 - Regularization
 - Dropout
- RNN
- LSTM
- GRU
- Midterm